

Questions with Answer Keys

MathonGo

Q1: 16 March (Shift 1) - Single Correct

Given below are two statements:

Statement I : H_2O_2 can act as both oxidising and reducing agent in basic medium.

Statement II : In the hydrogen economy, the energy is transmitted in the form of dihydrogen. In the light of the above statements, choose the correct answer from the options given below:

- (1) Both statement I and statement II are false
- (2) Both statement I and statement II are true
- (3) Statement I is true but statement II is false
- (4) Statement I is false but statement II is true

Q2: 16 March (Shift 2) - Single Correct

The correct statements about H_2O_2 are:

- (A) used in the treatment of effluents.
- (B) used as both oxidising and reducing agents.
- (C) the two hydroxyl groups lie in the same plane.
- (D) miscible with water.

Choose the correct answer from the options given below:

- (1) (A), (B), (C) and (D)
- (2) (A), (B) and (D) only
- (3) (B), (C) and (D) only
- (4) (A), (C) and (D) only

Q3: 17 March (Shift 1) - Single Correct

The INCORRECT statement(s) about heavy water is (are)

- (a) used as a moderator in nuclear reactor
- (b) obtained as a by-product in fertilizer industry.
- (c) used for the study of reaction mechanism

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(d) has a higher dielectric constant than water

Choose the correct answer from the options given below:

(1) (B) only

(2) (C) only

(3) (D) only

(4) (B) and (D) only

Q4: 18 March (Shift 2) - Single Correct

In basic medium, H_2O_2 exhibits which of the following reactions?

(A) $\text{Mn}^{2+} \rightarrow \text{Mn}^{4+}$

(B) $\text{I}_2 \rightarrow \text{I}^-$

(C) $\text{PbS} \rightarrow \text{PbSO}_4$

Choose the most appropriate answer from the options given below:

(1) (A), (C) only

(2) (A) only

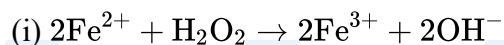
(3) (B) only

(4) (A), (B) only

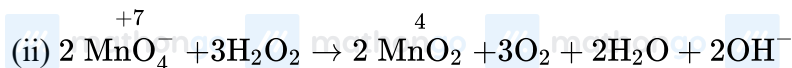
Answer Key**Q1 (2)****Q2 (2)****Q3 (3)****Q4 (4)**

Q1 (2)

(a) H_2O_2 can act as both oxidising and reducing agent in basic medium.



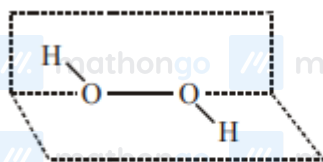
In this reaction, H_2O_2 acts as oxidising agent.



In this reaction, H_2O_2 acts as reducing agent.

(b) The basic principle of hydrogen economy is the transportation and storage of energy in the form of liquids or gaseous dihydrogen. Advantage of hydrogen economy is that energy is transmitted in the form of dihydrogen and not as electric power.

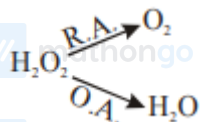
Q2 (2)



Structure of H_2O_2

(Open book type) $\rightarrow$ Non planar

H_2O_2 is used in the treatment of effluents.



act as both O.A. & R.A.

H_2O_2 is miscible in water due to hydrogen bonding.

Q3 (3)

The dielectric constant of H_2O is greater than heavy water.

Q4 (4)

In basic medium, oxidising action of H_2O_2 $\text{Mn}^{2+} + \text{H}_2\text{O}_2 \rightarrow \text{Mn}^{4+} + 2\text{OH}^-$

In basic medium, reducing action of H_2O_2 $\text{I}_2 + \text{H}_2\text{O}_2 + 2\text{OH}^- \rightarrow 2\text{I}^- + 2\text{H}_2\text{O} + \text{O}_2$

In acidic medium, oxidising action of H_2O_2 $\text{PbS(s)} + 4\text{H}_2\text{O}_2(\text{aq}) \rightarrow \text{PbSO}_4(\text{s}) + 4\text{H}_2\text{O}(\ell)$

Hence correct option (d)